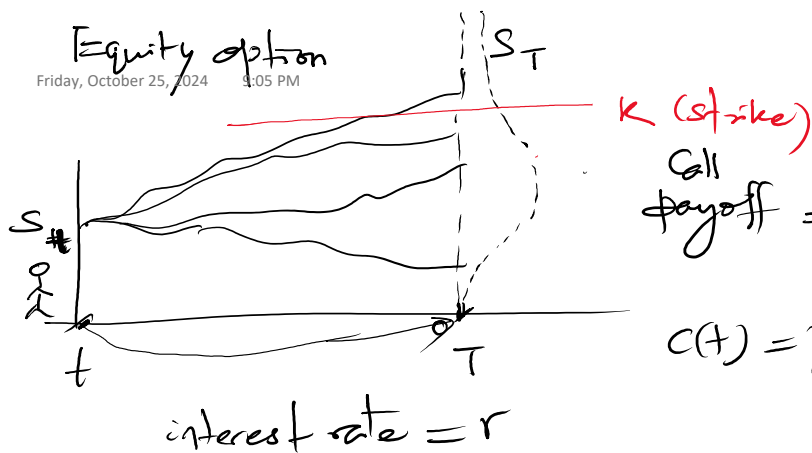




$$\text{put payoff} = (K - S_T)^+$$

$$\text{call payoff} = (S_T - K)^+ = \max(S_T - K, 0)$$

$$C(t) = ? \quad \mathbb{E}[\text{discounted payoff}]$$

$$= \mathbb{E}\left[e^{-r(T-t)} \times (S_T - K)^+\right]$$

$$C(t) = S_t N(d_1) - K e^{-r(T-t)} N(d_2) \rightarrow \text{BSM}$$

$\equiv$
 C_{mult}
 $\uparrow$

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

$$P(t) = -S_t N(-d_1) + K e^{-r(T-t)} N(-d_2)$$

$\underline{S_t} \rightarrow F(t, T) = \underline{F_t}$ $\mathbb{E}_t[L(T_1, T_2)] = F(t, T_1, T_2)$
 $\underline{F(t, T)}$ $\mathbb{E}_t^Q[S_T] = \underline{F(t, T)} = \underline{F_t}$
 $\underline{F(T, T)} = S_T$ $\underline{S_t} e^{r(T-t)}$

$$C(t) = e^{-r(T-t)} \cdot (F_t N(d_1) - k N(d_2)) \rightarrow \text{Black 76}$$

$$= \underline{F_t \cdot e^{-r(T-t)} \cdot N(d_1)} - \underline{k e^{-r(T-t)} N(d_2)}$$

$$= \underline{S_t N(d_1) - k e^{-r(T-t)} N(d_2)}$$

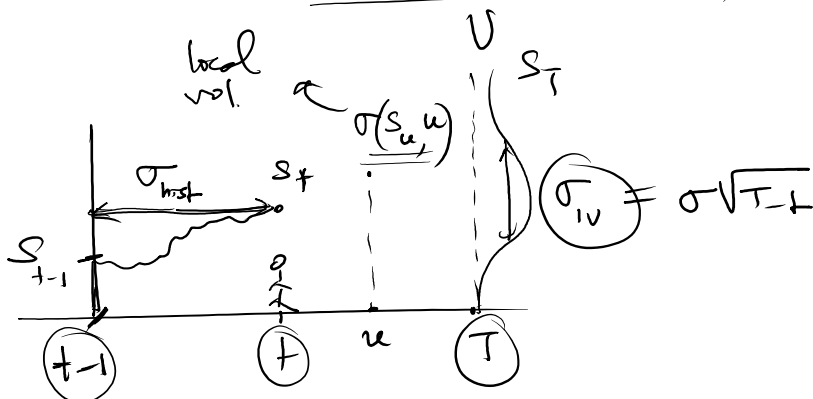
$$d_1 = \frac{\ln(F_t/k) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

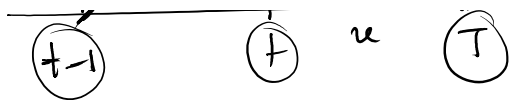
$$= \frac{\ln\left(\frac{S_t e^{r(T-t)}}{k}\right) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}} = \frac{\ln(S_t/k) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

annulized vol. $\rightarrow$ Terminal volatility
 $\sigma\sqrt{T-t} = v$



$$d_1 = \frac{\ln(F_t/k) + \frac{1}{2}v^2}{v}, \quad d_2 = \frac{\ln(F_t/k) - \frac{1}{2}v^2}{v}$$





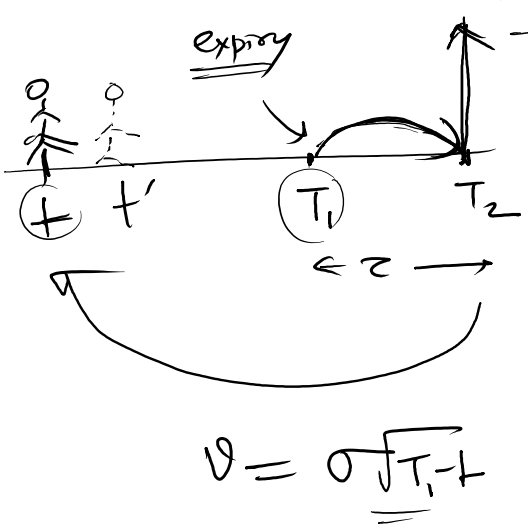
Interest rate Capslets / Floorlets

Friday, October 25, 2024 10:11 PM

Capslet = Call

Floorlet = put

$$(S_T - k)^+ \rightarrow$$



$$\frac{N\tau (\underline{L(T_1, T_2)} - \underline{k})^+}{\underline{\quad}} = \underline{N\tau} \max(\underline{L(T_1, T_2)}^f - k, 0)$$

$$\text{Capslet}(t) = \underline{N\tau} P(t, T_2)^f \left(\underline{F(t, T_1, T_2)}^f N(d_1) - k N(d_2) \right)$$

$$d_1 = \frac{\ln(F_t/k) + \frac{1}{2}\sigma^2 \tau}{\sigma \sqrt{\tau}}$$

$$d_2 = \frac{\ln(F_t/k) - \frac{1}{2}\sigma^2 \tau}{\sigma \sqrt{\tau}}$$

$$\text{Floorlet} \rightarrow \text{pay off} = N\tau (k - L(T_1, T_2))^+$$

$$\text{Floorlet}(t) = N\tau P(t, T_2) \left(-F(t, T_1, T_2) N(-d_1) + k N(-d_2) \right)$$

$$d_1(k, F, v) = \frac{\ln(F/k) + \frac{1}{2}v^2}{v}$$

$\omega = +1$, Gpleh
 -1 , Floorlel

$$d_2(k, F, v) = \frac{\ln(F/k) - \frac{1}{2}v^2}{v}, \quad v = \sigma \sqrt{T_1 - t}$$

$$B_1(k, F, v, \omega) = F \omega N(\omega d_1(k, F, v)) - k \omega N(\omega d_2(k, F, v))$$

$$V(t, T_1, T_2, \tau, F, k, \sigma) = \overline{(N\tau)} P(t, T_2) B_1(k, F, v, \omega)$$

Caps/Floors.

Friday, October 25, 2024

10:29 PM

Cap = series of Caplets

Floor = series of Floorlets